# Detecting Order-Dependent Flaky Tests in Java Projects

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Flaky tests pass and fail on the same code. The detector of this paper reruns each test in shuffled orders and labels 51 of 533 tests as flaky.

CCS Concepts: • **Software and its engineering → Software testing and debugging**.

Additional Key Words and Phrases: flaky tests, test order dependence, regression testing

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## 1 Introduction

Flaky tests pass and fail on the same code without any change to the program under test. Such tests waste the time of developers and hide real faults. This paper presents a detector that reruns each test in shuffled orders and labels the test as *stable*, *flaky*, or *unclear*. The detector runs as a plug-in of `mvn test` and reads the reports of the Surefire plug-in. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

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## 2 Approach

For a test $t$ and an order $o$, the detector records the outcome $r(t, o) \in \{pass, fail\}$. A test is flaky when two orders give different outcomes, and the detector stops after $n$ orders with

$$\exists\, o_1, o_2 \colon r(t, o_1) \neq r(t, o_2), \qquad \sum_{i=1}^{n} \frac{x_i^2}{\sqrt{n}} \leq \alpha. \tag{1}$$

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Table 1. Tests per label

| Label | Tests | Percent |
|---|---|---|
| stable | 450 | 84.4 |
| flaky | 51 | 9.6 |
| unclear | 32 | 6.0 |

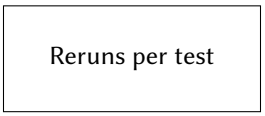



Fig. 1. The detector reruns each test in shuffled orders.

```
@Test
public void testCache() {
    assertEquals(3, cache.size());
}
```

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## 3 Evaluation

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[1]The detector and its data are available to the reviewers.

## 4 Conclusion

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